Assignment 4

DAY2: Routers and Switches Theory (30 min): Basic Function and differences between routers and switches. Practical (30 min): Identify port and LEDs on a router and switch.

Task 1: Labelled Diagram of Router and Switch

A. Typical Home Wi-Fi Router

A home Wi-Fi router has different ports and indicators used for connecting the Internet and local devices.

Important parts:

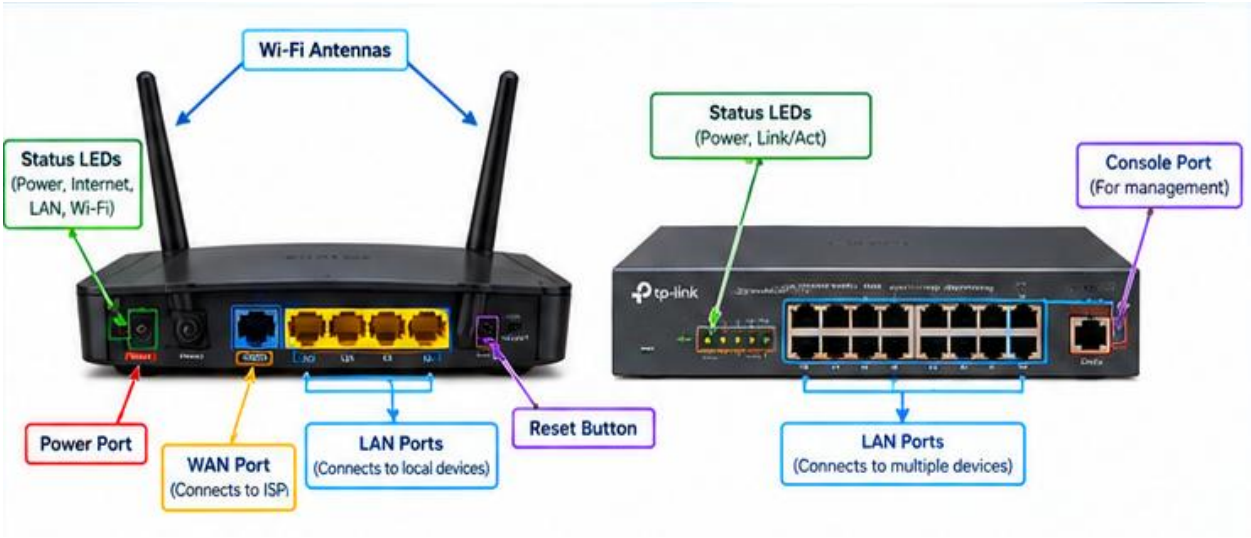
- **Power Port:** Used to connect the router's power adapter.
- **WAN/Internet Port:** Connects the router to the modem or ISP equipment.
- **LAN Ports:** Used to connect computers, TVs, switches and other wired devices.
- **Status LEDs:** Show the operating status of power, Internet/WAN, LAN and Wi-Fi.
- **Reset Button:** Used to reset the router to its default configuration.

B. Network Switch

A network switch is used to connect multiple wired devices within a Local Area Network (LAN).

Important parts:

- **Power Port:** Provides electrical power to the switch.
- **Ethernet/LAN Ports:** Used to connect computers, printers, servers and other network devices.
- **Status LEDs:** Show power and link/activity status for the ports.
- **Console Port:** Available on many managed switches for configuration and management.



Task 2: Three Key Differences Between Router and Switch

Feature	Router	Switch
1. Main Function	Connects different networks and routes packets between them.	Connects multiple devices within the same local network.
2. Address Used	Mainly uses **IP addresses** to determine where packets should go.	Mainly uses **MAC addresses** to forward Ethernet frames to the correct device.
3. Network Connection	Can connect a LAN to another network such as the Internet/WAN.	Primarily connects devices within the same LAN.

Explanation

A **router** works mainly at **Layer 3 (Network Layer)** and makes forwarding decisions using IP addressing.

A **switch** works mainly at **Layer 2 (Data Link Layer)** and uses MAC addresses to forward Ethernet frames within a LAN.

Task 3: Identify Ports and LEDs

A. Router

A typical home router contains the following ports and indicators:

1. Power Port

The power adapter is connected here to provide electricity to the router.

2. WAN/Internet Port

This port is used to connect the router to the modem or ISP/network connection.

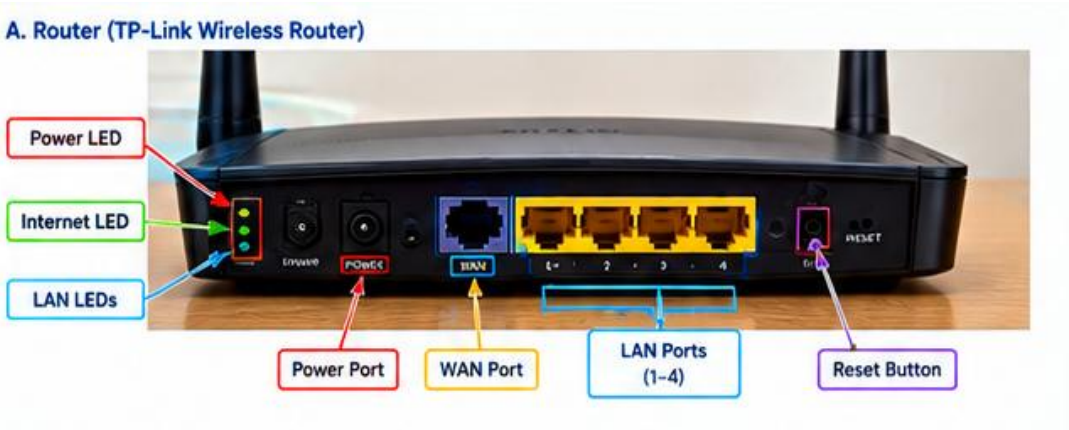
3. LAN Ports

LAN ports are used to connect wired devices such as computers, TVs, printers or switches.

4. LED Indicators

LEDs may indicate:

- Power status
- Internet/WAN status
- LAN link/activity
- Wi-Fi status



B. Network Switch

1. Power Port

Used to connect the switch to its power adapter or power supply.

2. Ethernet Ports

Used to connect multiple wired devices to the LAN.

3. LED Indicators

Port LEDs normally indicate whether a network link is active and whether data is being transmitted.

4. Console Port

Some **managed switches** have a console port used for configuration and management.



Task 4: Gaming Café Network Scenario

Scenario

Imagine that I am setting up a small gaming café where customers need a stable network for **IPL streaming and multiplayer gaming**. There are **8 gaming PCs** that need to connect to the network.

Network Setup

The **ISP line should be connected to the router first**. The router will provide the connection between the Internet and the local gaming network.

The router is then connected to a **network switch** using an Ethernet cable. The eight gaming PCs are connected to the switch using separate Ethernet cables.

Network Flow

ISP/Internet
↓
Router
↓
Network Switch
↓
PC 1
PC 2
PC 3
PC 4
PC 5
PC 6
PC 7
PC 8

Step-by-Step Setup

Step 1: Connect the ISP/modem connection to the router's **WAN/Internet port**.

Step 2: Configure the router to provide the Internet connection and local network services.

Step 3: Connect one LAN port of the router to the **uplink/LAN port of the switch**.

Step 4: Connect all **8 gaming PCs** to the switch using Ethernet cables.

Step 5: Test the network connection on each PC and confirm that all systems can communicate with the network and access the Internet.

Justification

The **router is connected first** because it connects the local network to the ISP/Internet and performs functions such as IP routing and network address translation.

The **switch is used after the router** because it provides multiple Ethernet ports for connecting the 8 gaming PCs to the same local network.

Using Ethernet connections for gaming PCs provides a stable wired connection and avoids relying on Wi-Fi for the main gaming systems.

